

PORTFOLIO · 2026

Nick Lian

I take ideas to shipped software,
alone, with AI agents — and set the
rules that keep them from breaking.

UBC Sauder, Class of 2026 · Finance + Business Analytics

Open to anywhere in Canada

nicklian0315@gmail.com github.com/NickkkLian

SIX MONTHS, ONE PERSON

Four kinds of things I built

The numbers are output, not users, as of Sept 2026.

42

projects

974

logged updates

25

apps on one private database

29

public repos

Delivered to others

- Letterkeep
Reference-letter system · in use by a professor
- Query Mirror
SQL teaching explorer · no-code tool for a professor

Public

- Scholar Outflow Lab
13 countries, recomputed weekly
- CallDelta
Earnings-call sentiment · tone → residual signal
- Vacancy Signal
Ghost-job trading signal · 310K postings

Used every day

- 25-app personal operating system

Runs on its own

- Multi-agent content pipeline
topic to release pack
- 10 scheduled jobs
- Guardrails and sentinels
machine blocks dangerous operations

What I built, and the decision inside each

WORK 1 · DELIVERED TO A PROFESSOR

Letterkeep Reference-letter system

Roster in, students screened by grade and attendance, documents uploaded, emails out. Shipped as a web app in a day; rebuilt as a desktop app after reading the privacy law, so student data stays on the professor's machine.

3 months · 56 iterations · Python / Flask / SQLite / OCR

THE POINT Nobody asked. I read the law and changed the architecture.

WORK 2 · USED EVERY DAY

A 25-app personal operating system

Single-file apps with no build step and one shared convention set, all on one private database. The website reads only data marked public and has zero access to the rest.

25 apps · 11 with built-in AI · count gate before every deploy

THE POINT Permission boundaries fixed in the architecture, not in habits.

WORK 3 · RUNS ON ITS OWN

Multi-agent content pipeline

Topic to release pack across several content channels. Agents exchange files, never chat; resumable state machine, bounded retries, a QC loop that sends failures back, and a human always publishes.

32 modules · 20 skills · usage ledger reconciled to the platform bill

THE POINT My own ledger covered 1.2% of the real bill. The bill became the source of truth.

WORK 4 · KNOWLEDGE

Structured knowledge bases built with AI

Thousands of entries across several private libraries. AI does the entry work; every entry carries a permanent ID and a source; reuse is by pointer, never by copy.

Verbatim first · no lossy simplification · checks before and after every change

THE POINT The hard part is the acceptance discipline, not the AI.

Six public projects

Each one is a complete, documented tool.

Query Mirror

No-code SQL teaching tool for a professor: click tables, filters and groups, the SQL appears live. Deterministic by design, no LLM.

Vanilla JS · sql.js (SQLite in WebAssembly) vendored
[NickkkLian/SQL-Viz-for-Edu](#)

Scholar Outflow Lab

Where researchers from 13 countries end up, from OpenAlex open data. Ranked by Wilson lower bound so small samples sink. Recomputed weekly.

Python stdlib scraper · GitHub Pages · launchd
[NickkkLian/Scholar-Outflow-Lab](#)

CallDelta

Management framing vs analyst pushback, by topic, across quarters. Does tone explain the residual move net of the EPS surprise?

Python · JSON schema + Pydantic · static dashboard
[NickkkLian/Earnings-Call-Sentiment-Analyser](#)

Vacancy Signal

Ghost postings detected in 310K listings, turned into a cross-sectional signal. Walk-forward backtest; +1.9pp/yr vs SPY, 2019–2026. Team of three.

XGBoost · GMM · SHAP · Revelio data
[NickkkLian/Ghost-Job-Detection-And-Trading-Signal](#)

Clipbind

Saved Xiaohongshu posts and Bilibili videos in one library; Claude merges selected notes into one writeup. Data stays in your browser and your own repo.

Single-page app · IndexedDB · MIT · live demo
[NickkkLian/content-organizer](#)

ApplyLedger

A multi-region job-search tracker in one HTML file: prompt generators, batch resume tailoring, and the GitHub Contents API as its backend.

React 18 in one HTML file · GitHub Contents API
[NickkkLian/Job-Tracker](#)

Ship first, then let machines keep it honest

Most guardrail rules come from a real incident. I don't hand-write code: I define the problem, set the constraints, verify the result, and own the result.

Keeping systems reliable

- A pre-tool hook blocks dangerous operations
force push, mass deletion, pushing while behind · 421 lines · ask by default
- Sentinels raise alarms
database shrinks, copies drift, sessions collide
- The reviewer is never the doer
an audit agent judges evidence only
- Three backup tiers, restores actually rehearsed

What I can do

- Take an idea to production alone
- Set the rules that keep systems reliable
- Python backends, single-file frontends, agent orchestration
AI sessions in parallel, one repo each
- Product judgment: what is worth building, what to cut
- Business grounding: valuation, financials, backtesting
bank valuation work · CFA Level I